



Poultry Value Chain Financing and Environmental Sustainability in South Africa: Evidence from a Mixed Methods Study



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Abstract: The poultry sector plays a critical role in food security, rural income generation, and economic development in South Africa. However, its rapid expansion has intensified environmental challenges such as waste accumulation, water contamination, greenhouse gas emissions, and pressure on natural resources. This study examines how poultry value chain financing influences environmental sustainability outcomes using a mixed methods approach. Primary data were collected from 45 respondents in Gauteng Province through structured questionnaires, complemented by 9 key informant interviews. Quantitative data were analysed using IBM SPSS version 26, while qualitative data were analysed using NVivo version 14. The results reveal a significant positive relationship between access to formal financing and the adoption of sustainable practices, including manure management, water efficiency, and energy-saving technologies. However, limited access to institutional credit constrains small-scale farmers, leading to continued reliance on environmentally harmful production methods. The study also highlights the role of governance frameworks and green financing mechanisms, including policy incentives, risk sharing instruments, and sustainability linked credit, in shaping environmental outcomes across the poultry value chain. The findings suggest that value chain financing plays an important role in promoting environmental sustainability and that targeted green financing instruments may facilitate the adoption of cleaner production systems. This study contributes empirical evidence to the growing discourse on sustainable agri-food systems and provides policy recommendations for strengthening environmentally responsible financing in the poultry sector.

Keywords: Agricultural value chains; Environmental; Poultry sector; Sustainability; Value chain financing

1 Introduction

Agricultural value chain finance has been widely recognized as a critical mechanism for enhancing productivity, market integration, and inclusive growth in developing economies [1]. By facilitating access to credit, input financing, and structured market linkages, value chain financing enables farmers to overcome liquidity constraints and invest in improved production systems [2]. Previous studies highlight that financial inclusion within agricultural value chains not only strengthens economic performance but also supports technological adoption and risk management among smallholder producers [3].

Despite these benefits, agricultural intensification particularly in the livestock and poultry sectors has been associated with significant environmental challenges. Poultry production systems are known to generate substantial environmental pressures, including waste accumulation, water contamination, high energy consumption, and greenhouse gas emissions [4]. These environmental externalities are often exacerbated in contexts where production expansion is not matched by adequate investment in sustainable technologies and environmental management systems.

Emerging research increasingly emphasizes the relationship between financing and sustainability transitions in agricultural systems. Access to finance enables farmers to adopt environmentally sustainable practices such as efficient waste management, renewable energy use, and water conservation technologies [5]. In contrast, persistent

financial constraints often force producers to rely on low-cost and resource-intensive production methods, which contribute to environmental degradation and undermine long-term sustainability.

Beyond financial access, governance and institutional frameworks also influence the effectiveness of sustainability transitions. Policy incentives, environmental regulations, and sustainability-linked financial instruments play an important role in determining whether financing mechanisms encourage green investments or reinforce unsustainable production practices [6]. Weak governance structures and limited availability of green finance instruments continue to constrain small-scale farmers' ability to adopt environmentally responsible production systems [7].

However, existing studies have largely examined agricultural finance and environmental sustainability in isolation, with limited empirical evidence on how value chain financing directly shapes environmental outcomes in the poultry sector particularly in developing economies such as South Africa [8]. Furthermore, prior research has predominantly focused on productivity and market integration, with insufficient attention to the environmental implications of financing structures.

More importantly, existing studies lack empirical validation of the causal pathways linking financing mechanisms to measurable environmental outcomes within poultry value chains, particularly when considering governance and institutional influences [9]. This study addresses the identified research gap by developing a conceptual framework linking poultry value chain financing and environmental sustainability outcomes. The framework is empirically tested under the influence of governance and regulatory systems (see Figure 1). The framework further emphasizes the moderating role of governance and green finance instruments, including regulatory enforcement and sustainability linked credit schemes, in strengthening the relationship between financing access and environmental sustainability outcomes [10]. It integrates insights from global value chain theory, institutional economics, and environmental sustainability perspectives to explain the mechanisms through which financing influences production behaviour, ecological performance, and risk management [11].

The main aim of this study is to examine the environmental implications of financial arrangements within the poultry value chain and to empirically assess how value chain financing influences environmental sustainability outcomes. Methodologically, the study adopts a mixed methods approach combining quantitative survey analysis and qualitative interviews to analyse the relationship between financing and sustainability outcomes. The findings indicate that financing instruments are not neutral economic tools but can either incentivize green transitions or reinforce unsustainable intensification depending on governance quality. The study contributes empirical evidence to the growing discourse on sustainable agri-food systems and supports decision-makers in designing financing frameworks that promote both industry growth and environmental protection in South Africa.

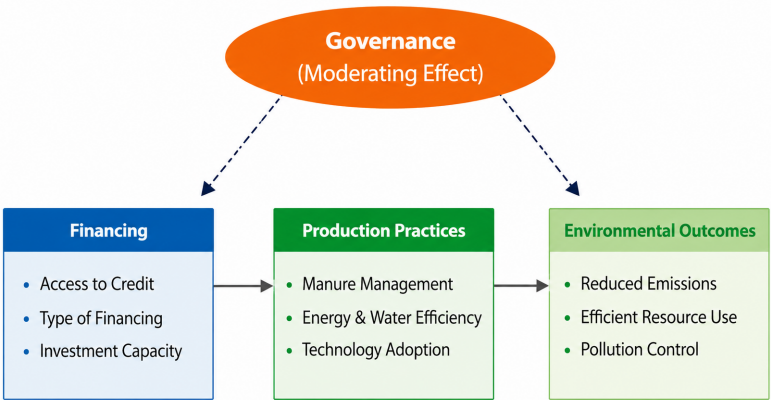


Figure 1. Conceptual framework linking value chain financing and environmental sustainability

2 Methodology

This study adopted a mixed-methods approach design to examine how poultry value chain financing influenced environmental sustainability in South Africa. The approach integrates quantitative survey data with qualitative interview insights to provide a comprehensive understanding of financing mechanisms and environmental outcomes. The conceptual framework developed in this study guides the empirical analysis by identifying key relationships between financing access, governance factors, and sustainability practices.

2.1 Research Design

The study adopted a mixed-methods approach design to examine how poultry value chain financing influences environmental sustainability in South Africa. Quantitative survey data and qualitative interview insights were

integrated to provide a comprehensive understanding of financing mechanisms and environmental outcomes.

2.2 Study Area and Context

The study was conducted in Gauteng Province, South Africa, where quantitative data were collected through structured questionnaires and qualitative data were obtained through semi-structured interviews. The data were analysed using IBM SPSS Statistics version 26 and NVivo version 14 to evaluate the relationship between value chain financing and environmental sustainability outcomes

2.3 Target Population and Sampling

The study utilized a mixed sampling approach. For the quantitative component, A total of 45 respondents were selected across different nodes of the poultry value chain using a random sampling technique. These included farmers, financial institutions, input suppliers, and other stakeholders.

For the qualitative component, 9 key informants were purposively selected based on their expertise and involvement in poultry value chain financing and environmental management. Although the sample size is relatively small, it is consistent with exploratory mixed-methods approach and reflects the challenges associated with accessing specialized stakeholders within agricultural value chains. The sample size is also adequate for regression analysis in exploratory studies.

This study adopts a value chain perspective, where multiple stakeholders (farmers, financial institutions, suppliers, and public actors) jointly influence financing flows and environmental outcomes. Therefore, combining these actors within a single analytical framework allows for a system-level understanding of sustainability dynamics. However, differences across stakeholder groups may influence perceptions of financing access and sustainability practices. For example, financial institutions may report systemic constraints, while farmers reflect operational realities. This heterogeneity may affect coefficient estimates and should be interpreted with caution.

2.4 Data Collection Methods

Quantitative data were collected through structured questionnaires administered to 45 respondents (26 males and 19 females), while qualitative data were obtained through semi-structured interviews with 9 participants, including representatives from public institutions, poultry farmers, and farmers' associations or farmers' unions.

2.5 Measurement of Variables

Environmental sustainability was measured using indicators such as waste reduction practices, water and energy efficiency, and compliance with environmental standards. Value chain financing variables included access to credit, supplier financing, microfinance products, and investment in green technologies. The measurement items were adapted from prior studies on agricultural finance and environmental sustainability, with modifications to fit the poultry value chain context. Responses were captured using a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). Environmental sustainability indicators included manure management (3 items), water efficiency (3 items), energy use (2 items), and renewable energy adoption (2 items). Financing variables included access to credit, type of financing, and investment capacity. The internal consistency of the measurement constructs was assessed using Cronbach's alpha. All constructs exceeded the acceptable threshold of 0.70, indicating satisfactory reliability. Composite indices for environmental sustainability and financing constructs were computed by averaging the respective Likert-scale items. Higher scores indicate greater adoption of sustainable practices and improved access to financing.

2.6 Data Analysis Techniques

Quantitative data were analysed using IBM SPSS Statistics version 26, employing descriptive statistics and regression analysis.

Qualitative data were analysed using NVivo version 14, applying thematic analysis to identify key patterns and insights related to financing and environmental sustainability.

In this study, Ordinary Least Squares (OLS) regression was employed to examine the relationship between value chain financing and environmental sustainability outcomes. The regression model is specified as follows:

$$\text{Environmental sustainability} = \beta_0 + \beta_1 (\text{Credit access}) + \beta_2 (\text{Financing type}) + \beta_3 (\text{Investment capacity}) + \sum \beta_k (\text{Control variables}) + \varepsilon$$

where, β_0 represents the intercept term, β_1 – β_k represent regression coefficients, and ε denotes the error term.

Control variables were coded as follows: firm size was measured based on operational scale (small, medium, large), experience was measured in years, and type of operation was treated as a categorical variable representing different poultry activities. These variables were included based on prior studies demonstrating that firm size, experience, and type of operation influence both access to finance and environmental performance.

3 Results

3.1 Descriptive Characteristics of Respondents

A total of 45 valid quantitative responses were analyzed from poultry sector stakeholders in Gauteng Province, South Africa. The sample comprised 26 male (57.78%) and 19 female (42.22%) respondents, indicating a moderate gender balance within the sector.

The demographic profile shows that the majority of respondents fall within the 25–40 age group (46.67%), followed by those aged 41–55 years (33.33%), suggesting that the poultry sector is largely driven by economically active individuals. In terms of racial composition, Black respondents constituted 51.11%, followed by White respondents (44.44%), while a small proportion (4.44%) identified as Coloured.

Educational attainment levels indicate a relatively skilled respondent base, with 48.88% possessing university or postgraduate qualifications, reflecting a growing level of professionalism and technical awareness within the poultry value chain.

From an operational perspective, the sample includes a diverse range of stakeholders across the poultry value chain. Retailers and wholesalers (17.78%), financial institutions (13.33%), and public institutions (11.11%) represent significant proportions, alongside poultry farmers engaged in broiler, layer, or mixed production systems. In terms of experience, most respondents (37.78%) have 4–7 years of service, followed by 33.33% with 1–3 years, indicating a relatively developing but increasingly established sector.

These findings suggest that the poultry value chain in Gauteng is characterized by diverse stakeholder participation, moderate experience levels, and a relatively educated workforce, although structural and financial constraints remain evident.

Table 1 presents the demographic and operational profile of respondents, highlighting the diversity of stakeholders across the poultry value chain, including farmers, financial institutions, input suppliers, and public sector actors. This distribution reflects the multi actor nature of the sector and provides a comprehensive basis for analysing financing and environmental sustainability dynamics.

Table 1. Operational and demographic profile of poultry sector stakeholders in South Africa

Demographic Characteristics	Category	Frequency (<i>n</i> = 45)	Percentage (%)
Age	Under 25	4	8.89
	25–40	21	46.67
	41–55	15	33.33
	Above 55	5	11.11
Gender	Male	26	57.78
	Female	19	42.22
Race	Black	23	51.11
	White	20	44.44
	Coloured	2	4.44
	Primary	1	2.22
Level of education	Secondary	5	11.11
	Certificate	7	15.56
	Diploma	10	22.22
	University degree	11	24.44
	Postgraduate education	11	24.44
	Broiler chickens farmer	7	15.56
Stakeholder type	Layers chickens farmer	3	6.67
	Both broiler & layers	4	8.89
	Hatchery	3	6.67
	Input supplier	4	8.89
	Financial institution	6	13.33
	Retailer & wholesaler	8	17.78
	Public institution	5	11.11
	Farmers union & associations	5	11.11
	Below 1 year	1	2.22
	1–3 years	15	33.33
Years in service	4–7 years	17	37.78
	Above 7 years	12	26.67
	Manager	10	22.22
Designation/rank	Director/owner	13	28.89
	Employee	22	48.89

3.2 Access to Value Chain Financing

Findings indicate that access to value chain financing remains uneven among poultry stakeholders in Gauteng Province. A majority of respondents (71%) depend on informal sources of finance, including personal savings, family contributions, and social lending groups, while only 29% have access to institutional financing such as bank loans, microcredit, or supplier credit.

Several structural barriers continue to constrain access to formal financing. These include stringent collateral requirements, lack of formal credit histories, limited financial literacy, and high risk perceptions associated with poultry farming due to disease outbreaks and market volatility.

Figure 2 shows that informal financing remains dominant among poultry actors, limiting investment in sustainable production technologies. This imbalance limits investment in environmentally sustainable production systems.

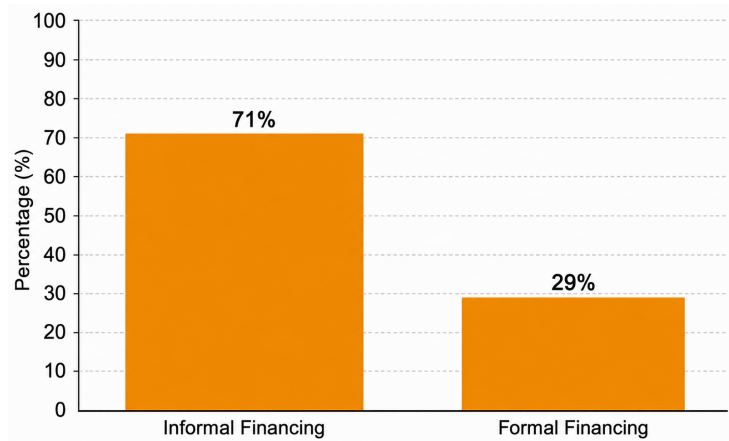


Figure 2. Types of financing accessed by poultry actors

3.3 Adoption of Environmental Sustainability Practices

Environmental sustainability performance was assessed using four key indicators: manure management, water recycling, energy efficiency, and renewable energy adoption. As presented in Table 2, adoption levels across all indicators remain critically low, with none exceeding 40%. Only 36% of poultry farmers have established proper manure management systems capable of preventing nutrient leakage and soil or groundwater contamination. Water recycling practices are adopted by just 27% of respondents, despite poultry production being highly water intensive and vulnerable to water scarcity risks. Energy efficient production systems including improved lighting and heating technologies are limited to 22% of enterprises, reflecting the sector's continued reliance on outdated, high emission equipment. The lowest adoption rate (11%) relates to renewable energy use, particularly solar power, which is considered a strategic solution for reducing carbon emissions and operational costs.

Table 2. Adoption of environmental sustainability practices in poultry enterprises

Sustainability Indicator	Adoption Level (%)
Proper manure management	36
Water recycling	27
Energy efficient lighting/heating	22
Renewable energy (solar)	11

The evidence suggests that low financial capacity and structural barriers to accessing capital constrain farmers' ability to integrate environmentally sound technologies. Small-scale producers, who constitute a substantial proportion of the industry, continue to rely on low-cost production methods that may contribute to environmental degradation, increasing risks of waste accumulation, greenhouse gas emissions, and water resource depletion. The weak uptake of solar energy demonstrates the absence of incentives and institutional support for clean energy transitions, further widening environmental sustainability gaps across the value chain. These findings highlight the urgent need for financing interventions specifically aimed at supporting environmental compliance and green technology diffusion.

3.4 Impact of Financing on Sustainability Outcomes

The regression analysis, conducted using IBM SPSS Statistics version 26, revealed a positive and statistically significant relationship ($p < 0.05$) between access to formal value chain financing and environmental sustainability

outcomes. The regression results are presented in Table 3.

Table 3. Regression results of the effect of financing on sustainability outcomes

Variables	Coefficient (β)	Std. Error	t-Value	p-Value
Constant	1.245	0.312	3.99	0.000
Access to credit	0.421	0.108	3.90	0.000
Type of financing	0.287	0.095	3.02	0.003
Investment capacity	0.356	0.121	2.94	0.004
Firm size (Control)	0.198	0.082	2.41	0.017
Experience (Control)	0.145	0.073	1.99	0.049
Type of operation (Control)	0.167	0.088	1.89	0.061

The regression results confirm that access to credit ($\beta = 0.421$, $p < 0.001$), type of financing ($\beta = 0.287$, $p = 0.003$), and investment capacity ($\beta = 0.356$, $p = 0.004$) have statistically significant positive effects on environmental sustainability outcomes. Control variables such as firm size and experience also show moderate influence.

The model demonstrates acceptable explanatory power ($R^2 = 0.52$; Adjusted $R^2 = 0.48$) and is statistically significant ($F = 14.67$, $p < 0.001$). Diagnostic tests confirm no multicollinearity issues ($VIF < 5$). Further diagnostic tests confirmed the normality and homoscedasticity of residuals, indicating that the assumptions of the regression model were not violated. Further diagnostic tests confirmed the normality and homoscedasticity of residuals.

Overall, the results suggest that value chain financing is strongly associated with improved sustainability outcomes and may support sustainability transitions within the poultry sector in Gauteng Province.

Figure 3 illustrates the relationship between financing access and sustainability performance. These findings further support the association between access to financing and enhanced environmental sustainability performance. Without adequate access to external capital, farmers remain constrained to low-cost production methods that often result in excessive waste generation, inefficient resource use, and increased environmental degradation, thereby limiting long term competitiveness and sector resilience.

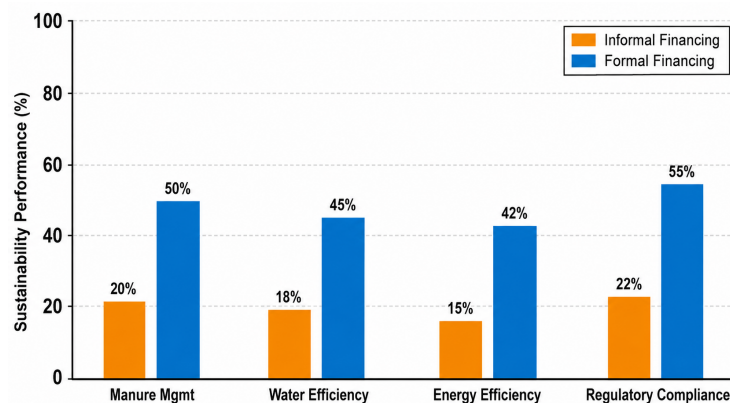


Figure 3. Relationship between financing access and sustainability performance

3.5 Qualitative Insights

Qualitative data from 9 participants, analysed using NVivo version 14, provided deeper insights into the financing–sustainability relationship. The profile of qualitative participants is presented in Table 4.

Table 4. Profile of qualitative participants ($n = 9$)

Stakeholder	Number	Data Collection Method
Public institutions	3	Semi-structured interviews
Poultry farmers	5	Semi-structured interviews
Poultry association/farmers' union	1	Semi-structured interviews
Total	9	

Key informants consistently highlighted several structural challenges. Financial institutions perceive poultry farming as high risk due to disease outbreaks, input price volatility, and market fluctuations, which limits credit

provision. Participants also emphasized the lack of targeted government support and green financing schemes to incentivize sustainable investments.

The high cost of environmentally friendly technologies such as waste management systems, solar energy, and water-efficient infrastructure was identified as a major barrier. Additionally, weak coordination among value chain actors increases operational inefficiencies and limits compliance with environmental standards.

These insights reinforce the quantitative findings, indicating that sustainability challenges are primarily driven by structural and financial constraints rather than a lack of awareness or willingness among stakeholders.

3.6 Summary of Key Findings

The findings demonstrate that the poultry value chain in Gauteng Province is constrained by limited access to formal financing, which significantly affects environmental sustainability outcomes.

Evidence from both quantitative and qualitative data shows that stakeholders with access to structured financing exhibit higher levels of environmental compliance and adoption of sustainable practices. In contrast, reliance on informal financing perpetuates low investment in environmentally efficient technologies.

Overall, the results highlight that improving environmental sustainability in the poultry sector is closely linked to strengthening financial inclusion, expanding targeted green financing mechanisms, and enhancing institutional and policy support frameworks.

4 Discussion

This study contributes to a deeper understanding of how poultry value chain financing is associated with environmental sustainability in South Africa, suggesting that financing plays an important role in shaping ecological outcomes. The poultry sector, particularly the small-scale production segment, continues to experience structural underinvestment constraints, where reliance on informal financing constrains the adoption of environmentally sustainable technologies [12]. This reinforces the argument that financial access serves as a gateway to sustainability, while financial exclusion perpetuates environmental degradation [13].

These results are consistent with prior studies showing that access to finance significantly enhances the adoption of sustainable agricultural practices and strengthens environmental compliance [14]. However, this study extends the existing literature by providing empirical evidence that financing constraints within the poultry sector not only limit productivity improvements but also intensify environmental degradation through continued reliance on resource inefficient and high emission production systems [15].

The findings are also aligned with broader evidence from agricultural finance literature, which emphasizes that improved access to credit enables farmers to invest in productivity enhancing and environmentally sustainable technologies [16]. Similarly, studies on contract farming and value chain participation suggest that structured financing arrangements can reduce financial constraints and facilitate the adoption of modern production practices, including environmentally efficient systems [17]. From a sustainability investment perspective, prior research highlights that financial inclusion plays a critical role in enabling firms to internalize environmental costs and adopt cleaner production processes.

In the African context, studies on poultry and agricultural systems have similarly identified limited access to finance as a key barrier to sustainable production, particularly among smallholder farmers who face structural constraints such as high credit risk and weak institutional support [18]. However, the present study advances this body of knowledge by demonstrating that financing constraints not only limit productivity but also directly exacerbate environmental degradation within the poultry value chain. This highlights the dual role of finance as both an economic enabler and an environmental determinant in emerging agri-food systems [19]. The empirical findings highlight clear disparities in sustainability behaviour between farmers with structured institutional financing and those operating without such support. Respondents accessing formal credit were significantly more likely to invest in improved manure management, water recycling systems, and energy efficient equipment [20]. This aligns with the theoretical expectation that financing influences production practices, which in turn shape environmental performance. In this context, environmental inefficiencies can be understood as downstream consequences of financing constraints within the value chain. Access to financing enables investment in environmental infrastructure such as waste management systems, water recycling technologies, and energy-efficient equipment. It also reduces liquidity constraints and supports the adoption of cleaner production technologies through supplier credit and sustainability-linked financing arrangements [21, 22].

The study also demonstrates that financial institutions' risk perceptions surrounding poultry farming characterized by disease vulnerability, price fluctuations, and high operational costs contribute to limited credit participation. These perceptions restrict lending to only a small portion of the value chain, typically medium scale and vertically integrated actors. Consequently, the sector evolves in an unbalanced manner, with small producers lacking both capital and incentives to comply with environmental regulations. The findings thus underscore the role of governance and institutional capacity in moderating finance–environment interactions. Weak enforcement of standards and absence

of green finance policies allow continued externalization of environmental costs, particularly in waste disposal and energy usage [23].

Another important insight concerns the structural characteristics of the poultry sector. As a growth oriented and densely concentrated agri food industry, rising production intensity increases risks of waste accumulation, groundwater contamination, and greenhouse gas emissions if sustainability investments do not keep pace [24]. The results therefore suggest a potential widening of the sustainability gap over time unless financing interventions proactively support cleaner production systems.

The findings also highlight the need for stronger institutional coordination. Policy implications derived from this study are substantial. Legislative pressure alone without matching financial support mechanisms is unlikely to deliver meaningful shifts in sustainability performance [25]. Integrated financial solutions are needed, such as sustainability-linked credit, performance-based subsidies for renewable energy adoption, and credit guarantee facilities that reduce lending risks for agricultural financial institutions. Similarly, stronger value chain collaboration, where lead firms co finance environmental upgrades for smallholders within supply agreements, could accelerate transition toward greener production [26].

Overall, the findings advance the thesis that environmental sustainability challenges within South Africa's poultry sector are fundamentally financial challenges. By reconfiguring financing models to reward and enable sustainable practices, the sector can improve its competitiveness, enhance regulatory compliance, and reduce environmental pressure. This empirical evidence, interpreted through the proposed conceptual framework, provides a foundation for policy makers, development finance institutions, and value chain leaders to design innovative financing solutions that balance poultry sector expansion with ecological responsibility [27].

This study contributes to the literature in three key ways. First, it provides empirical evidence linking value chain financing to environmental sustainability outcomes in the poultry sector. Second, it integrates financial and environmental perspectives within a unified analytical framework. Third, it highlights the moderating role of governance and institutional structures in shaping sustainability transitions.

Given the cross-sectional nature of the data, the findings should be interpreted as associations rather than causal relationships. The analysis is based on a relatively small sample of 45 quantitative respondents and 9 qualitative participants, and was conducted exclusively in Gauteng Province, South Africa. These differences across stakeholder groups should be considered when interpreting the results.

5 Conclusions

This study examined the relationships between access to value chain financing and the adoption of environmental sustainability practices in the poultry sector. The findings clearly demonstrate that financing plays a critical role in enabling farmers to transition from conventional production systems toward more resource efficient and environmentally responsible operations. Although poultry producers contribute significantly to food security and rural livelihoods, their capacity to invest in sustainable technologies remains severely constrained by financial limitations.

Results show that the majority of poultry farmers remain dependent on informal financing mechanisms, which are inadequate to support infrastructure upgrades required for effective waste management, water recycling, and renewable energy adoption. Low adoption rates of key sustainability practices indicate that environmental concerns are often deprioritized in favour of short-term cost reduction strategies. Farmers with access to formal financing showed significantly higher compliance with environmental standards and greater investment in production efficiency, confirming the hypothesis that financing functions as a key enabling factor in sustainability transitions.

The findings highlight the need for targeted green financing strategies to support environmentally sustainable poultry production. Expanding access to sustainability-linked loans, renewable energy financing, and risk-sharing mechanisms could improve environmental compliance while strengthening sector resilience.

This study is subject to several limitations. First, the relatively small sample size may limit the generalizability of the findings. Second, the inclusion of heterogeneous stakeholder groups may obscure differences in behaviour and access to financing across actors within the value chain. Third, the cross-sectional design restricts the ability to establish causal relationships between financing and environmental sustainability outcomes.

In conclusion, ensuring equitable access to value chain financing is central to driving a sustainable poultry production ecosystem. Strengthened financial mechanisms have the potential to improve environmental performance while enhancing the long-term competitiveness and resilience of poultry enterprises.

Author Contributions

Conceptualization, P.N. and G.M.; methodology, P.N. and G.M.; software, P.N.; validation, P.N. and G.M.; formal analysis, P.N. and G.M.; investigation, P.N.; resources, G.M.; data curation, P.N.; writing—original draft preparation, P.N.; writing—review and editing, P.N. and G.M.; visualization, P.N.; supervision, G.M.; project administration, P.N.; funding acquisition, G.M. All authors have read and agreed to the published version of the manuscript.

Ethical Approval

Ethical approval for this study was obtained from the Durban University of Technology Institutional Research Ethics Committee (DUT IREC) under clearance number IREC 249/24. Participation was voluntary, and all respondents provided informed consent. Confidentiality and anonymity were strictly maintained.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflict of interest.

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